Project Overview and Methodological Framework

1. Project Overview

This study investigates the impact of feature selection on classification accuracy in high-dimensional gene expression data. The central research question is:

“Does reducing the number of genes (features) improve or reduce classification performance?”

Gene expression datasets are characterized by:

- A very large number of features (thousands of genes)
- A relatively small number of samples

This imbalance leads to several challenges, including:

- Overfitting
- Increased computational cost
- Poor generalization of models

To address these issues, this project focuses on:

- Applying feature selection techniques
- Training classification models
- Comparing performance before and after feature reduction

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2. Experimental Workflow

Stage 1: Define Scope

To ensure clarity and feasibility, the study is limited to:

- 2–3 feature selection methods
- 2–3 classification algorithms
- 1–2 gene expression datasets

Example Configuration:

Feature Selection Methods:

- Filter Methods (e.g., Chi-square, Information Gain)
- Wrapper Methods (e.g., Recursive Feature Elimination - RFE)
- Embedded Methods (e.g., LASSO)

Classification Algorithms:

- Support Vector Machine (SVM)
- Random Forest
- k-Nearest Neighbors (k-NN)

Stage 2: Dataset Selection

Publicly available gene expression datasets will be used, such as:

- Leukemia dataset
- Colon cancer dataset
- Breast cancer dataset

These datasets can be sourced from:

- Kaggle
- UCI Machine Learning Repository
- Gene Expression Omnibus (GEO)

Stage 3: Data Preprocessing

Proper preprocessing is essential for reliable results. The following steps will be performed:

- Handling missing values
- Data normalization or standardization
- Encoding class labels (e.g., cancer vs normal)

Stage 4: Baseline Model Development

Before applying feature selection, baseline models will be trained using all available features.

Performance will be evaluated using:

- Accuracy
- Precision
- Recall
- F1-score

This serves as a control experiment for comparison.

Stage 5: Feature Selection and Model Training

Each feature selection method will be applied to reduce dimensionality. For each method:

- A subset of top-ranked features (e.g., top 50, 100, 200) will be selected
- Classification models will be retrained using the reduced feature set
- Performance metrics will be recorded

Stage 6: Performance Comparison

The results will be analyzed to answer the following:

- Does feature selection improve classification accuracy?
- Does it reduce computational cost?
- Which feature selection method performs best?
- What is the optimal number of features?

Results will be presented using:

- Comparative tables
- Graphs (e.g., accuracy vs number of features)

Stage 7: Analysis and Interpretation

Beyond presenting results, the study will provide detailed analysis, including:

- Reasons for improvements or declines in performance
- Stability and robustness of different methods
- Trade-offs between model accuracy and feature reduction

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3. Report Structure

Chapter 1: Introduction

- Background of gene expression analysis and machine learning
- Problem statement
- Aim and objectives
- Research questions
- Scope of the study

Chapter 2: Literature Review

- Challenges of high-dimensional data
- Overview of feature selection techniques:
- Filter methods
- Wrapper methods
- Embedded methods
- Review of related studies and findings

Chapter 3: Methodology

- Description of datasets used
- Data preprocessing techniques
- Feature selection methods applied
- Classification algorithms used
- Evaluation metrics

Chapter 4: Results and Discussion

- Baseline model performance
- Performance after feature selection
- Comparative analysis (tables and graphs)
- Interpretation and discussion of results

Chapter 5: Conclusion and Recommendations

- Summary of key findings
- Identification of best-performing methods
- Limitations of the study
- Suggestions for future research